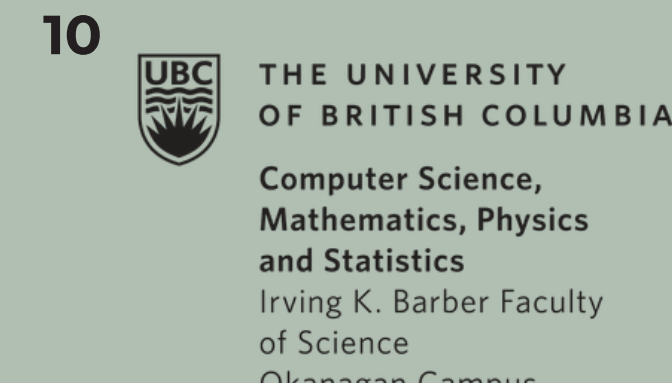


Agent-based Insights into Evolutionarily Stable Strategies in Mammalian Search Behaviour

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Background

Animals rely on movement to find food and mates while avoiding predators.¹ Research shows encounters increase with both speed and directional persistence,^{1,2} but higher speeds also raise energetic costs³, creating a trade-off between maximizing encounters and minimizing energy expenditure.

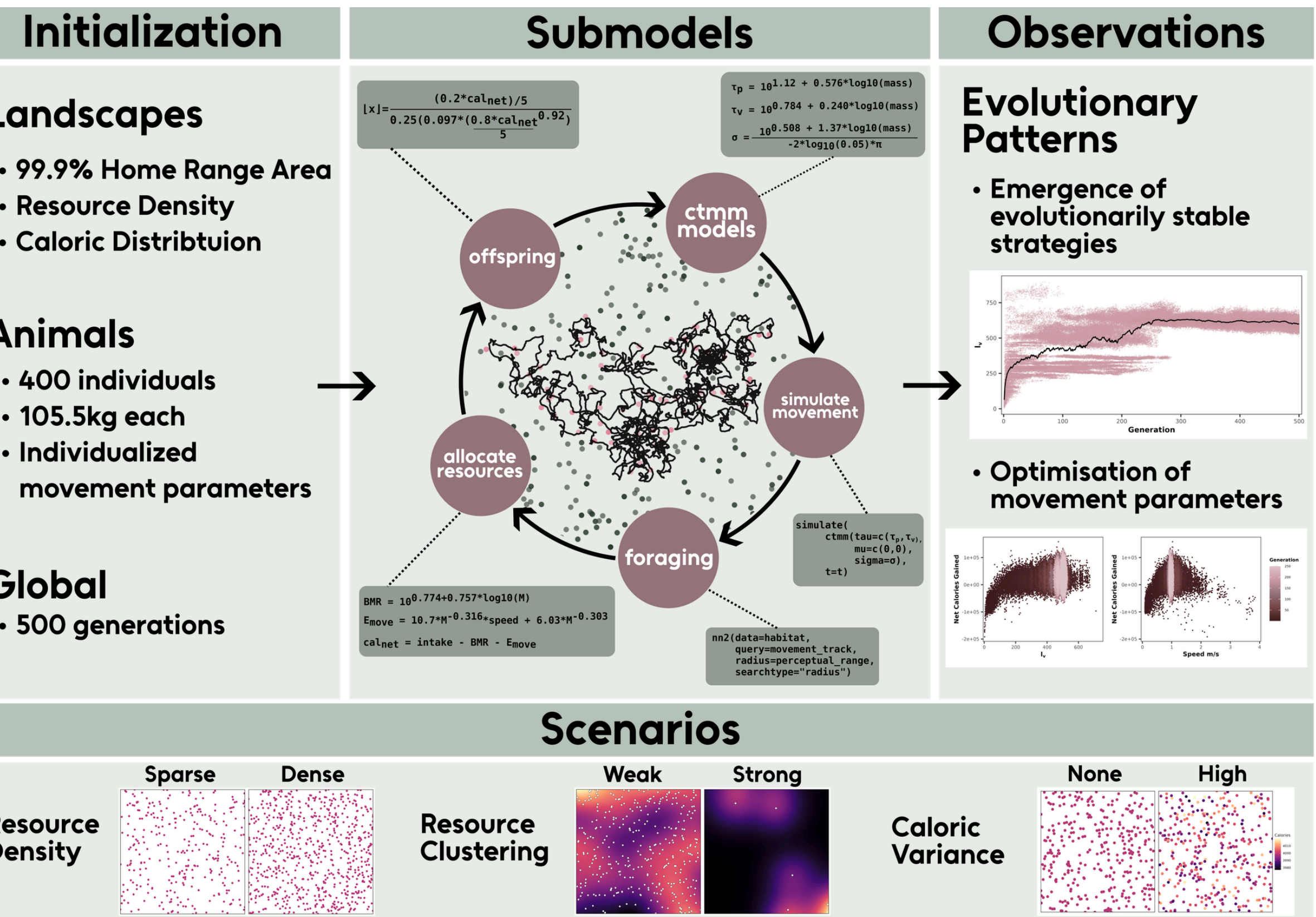
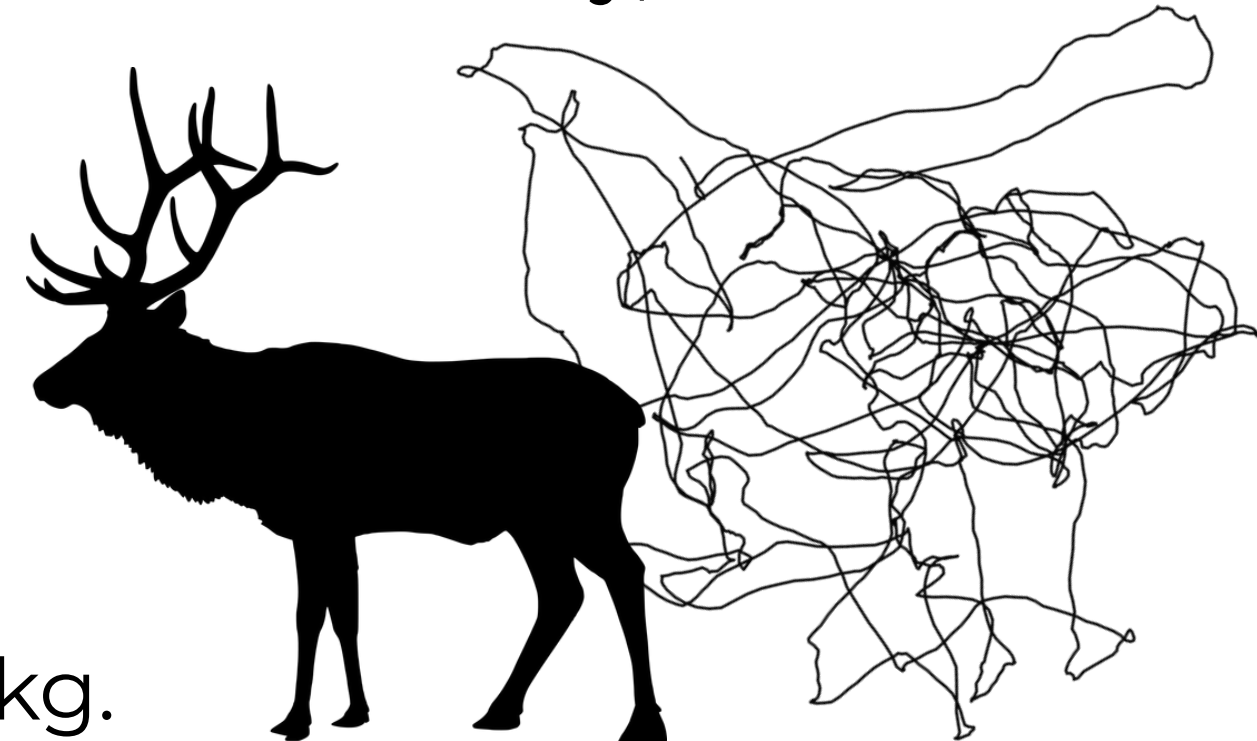
↑ speed + ↑ directional persistence >>> ↑ encounters

Fitness depends on foraging success, so movement strategies should balance these competing demands.⁴ Existing theory assumes movement is cost-free and predicts that animals facing bottom-up pressure should move more ballistically. We test if this holds when that assumption is removed.

Methods

A stochastic agent-based model was developed, parameterized by mammalian allometry, centered on terrestrial herbivores.

Simulations focused on populations of individuals with a body mass of 105.5 kg.



Energetic Costs Constrain Movement Strategies

Evolutionarily stable strategies emerge

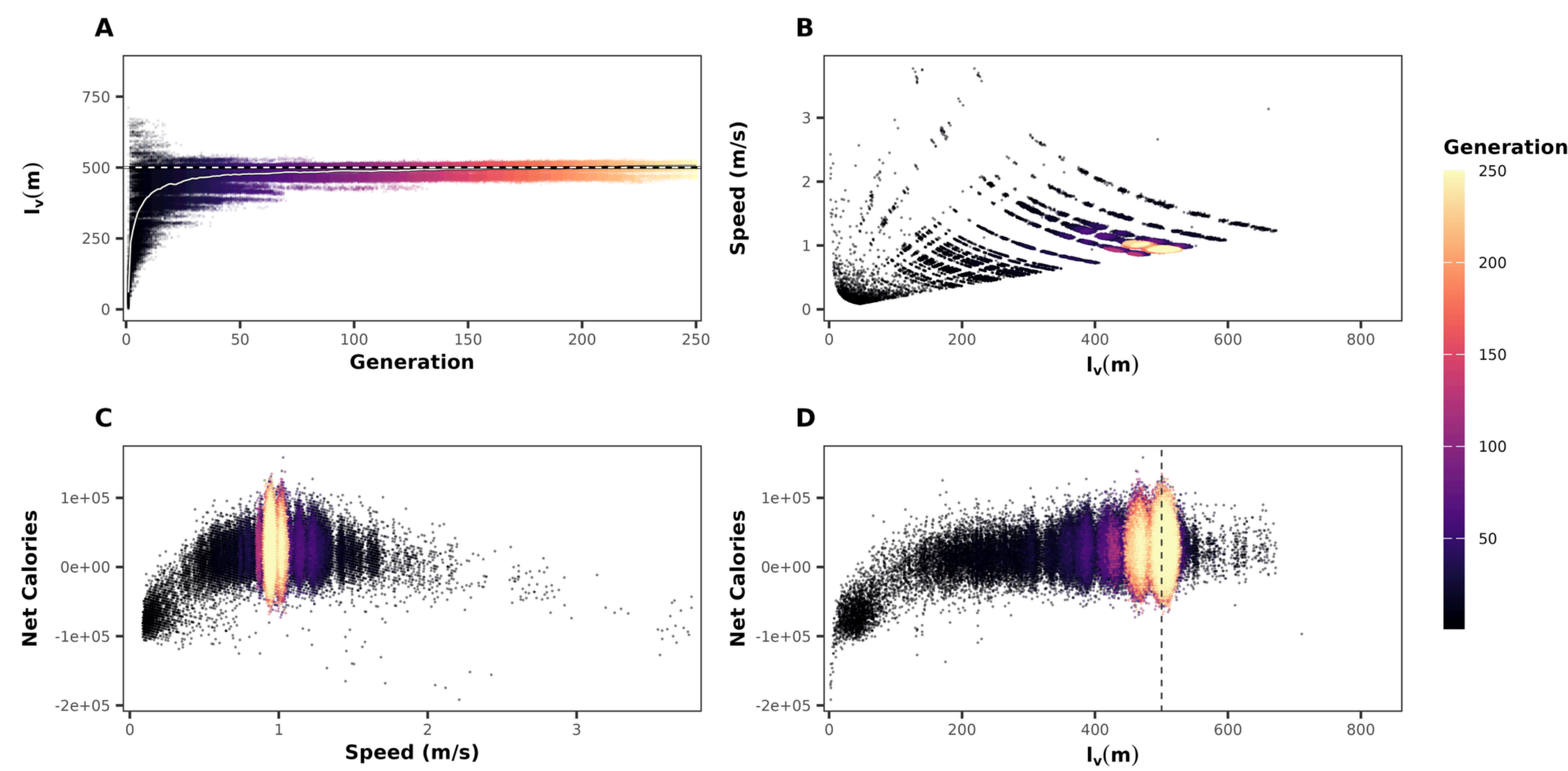


Figure 1: Evolution of movement strategy in a homogeneous landscape (250 patches, no clustering, no caloric variance). (A) Ballistic length scale (l_v) increases over generations (dashed line indicates stabilized value). (B) l_v and speed do not evolve independently. (C) Energy gained is optimized with respect to speed. (D) Energy gained is maximized at an intermediate, stabilized value of l_v . Points are coloured by generation. Evolutionarily stable strategies (ESS) involve the optimization of both traits.

Clustering selects for slow and diffusive movement

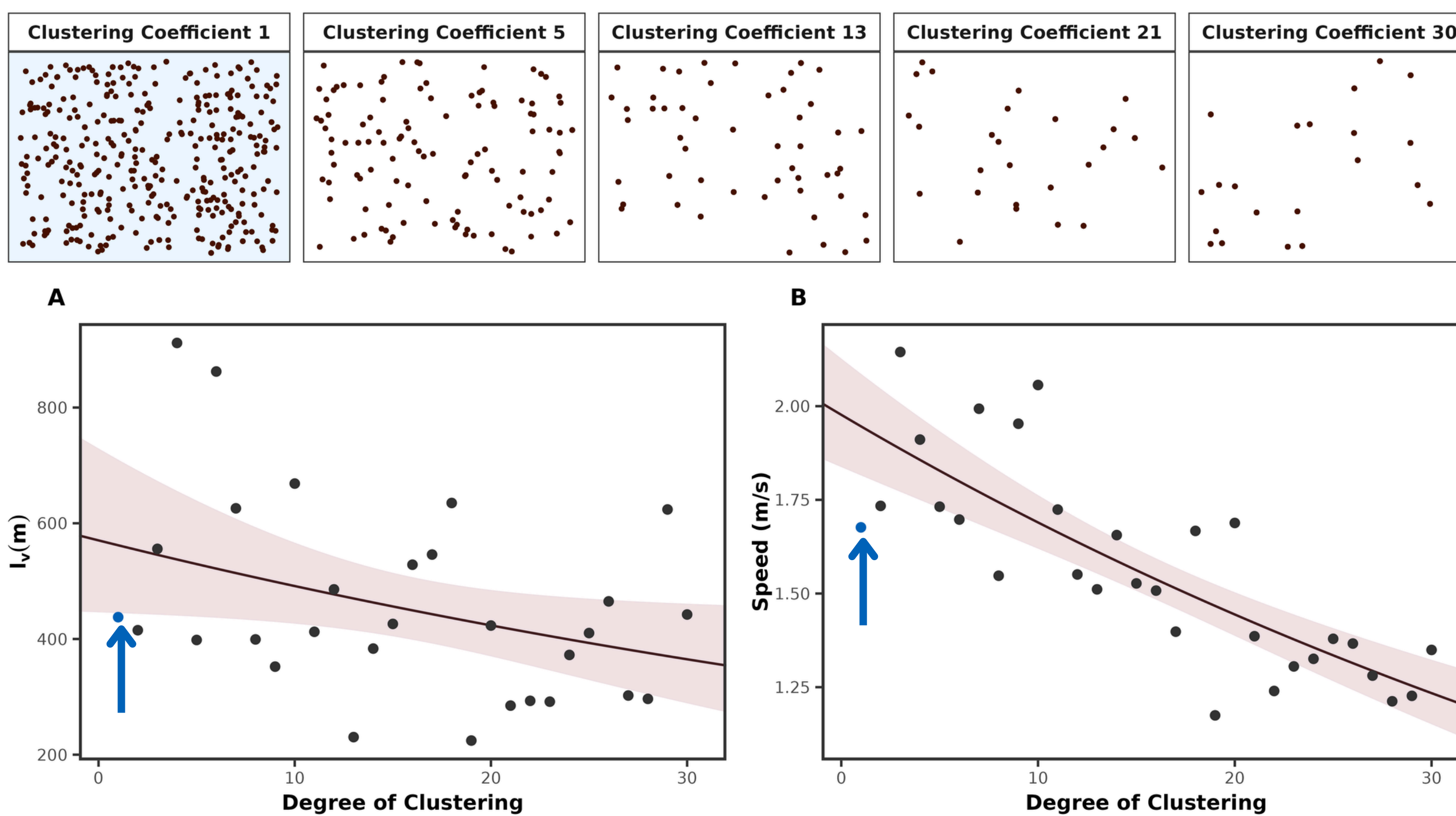


Figure 3: ESS in landscapes with increased clustering. (A) Stabilised ballistic length scale (l_v) decreases with resource clustering (Gamma GLM, log link: $\beta = -0.015$, $p = 0.0441$). (B) Stabilised speed (m/s) also decreases with resource clustering (Gamma GLM, log link: $\beta = -0.016$, $p < 0.001$). Blue arrow indicates parametrization that is consistent across all scenarios.

Future Work

- testing additional landscape scenarios
- adding predators to the system

References & Code Availability

All code used in the project is publicly available on GitHub. Scan the QR code to view the repository and references.



Acknowledgements



Increasing resource density selects for faster speeds

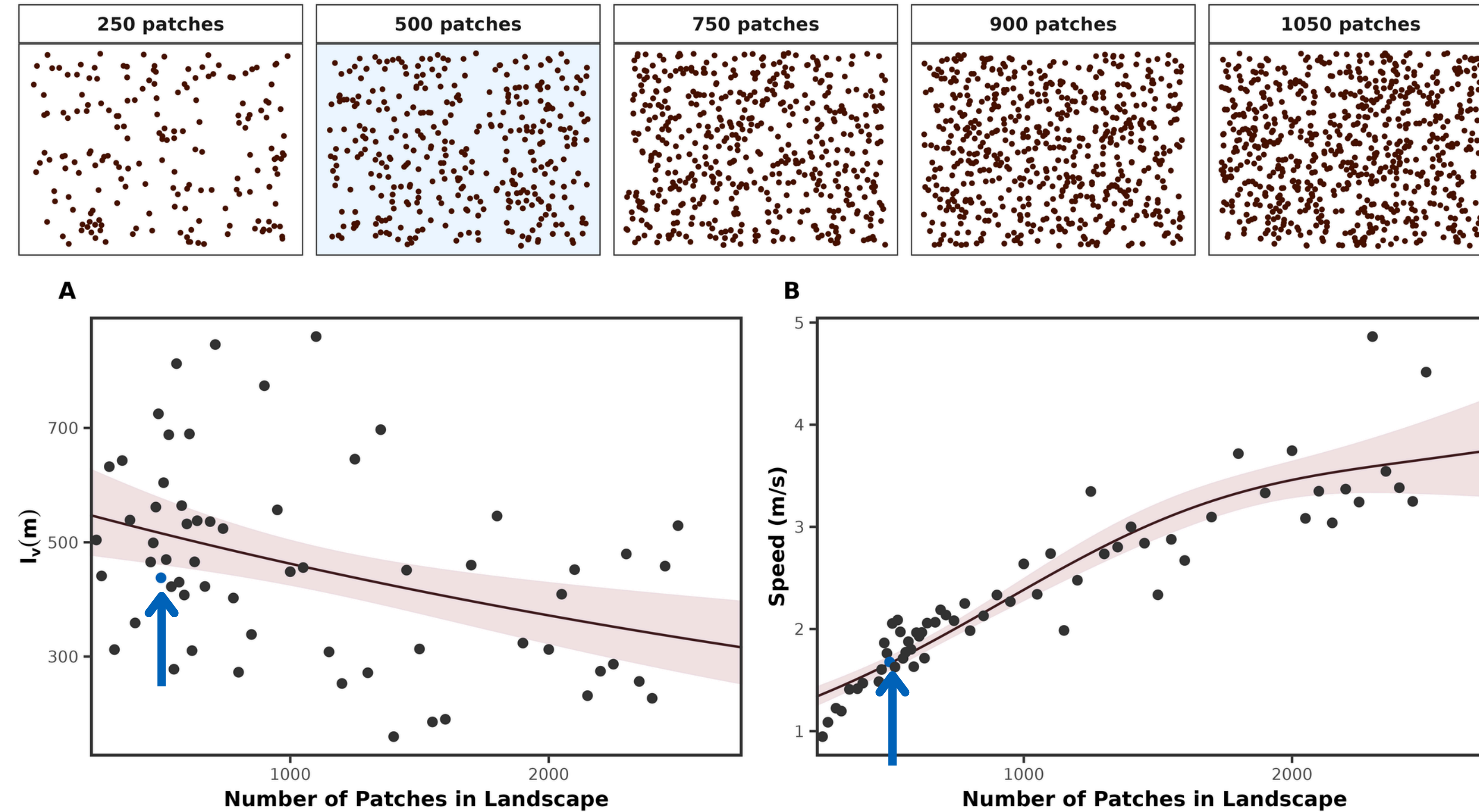


Figure 2: ESS in homogeneous landscapes with increasing density. (A) Stabilised ballistic length scale (l_v) does not show a clear relationship with resource density (Gamma GLM, log link: $\beta = 2.14 \times 10^{-5}$, $p = 0.889$). (B) Stabilised speed (m/s) increases with resource density to a saturation point (Gamma GAM, log link: $\text{edf} = 1.97$, $F = 107$, $p < 0.001$). Blue arrow indicates parametrization that is consistent across all scenarios.

Unpredictability selects for faster, more ballistic movement

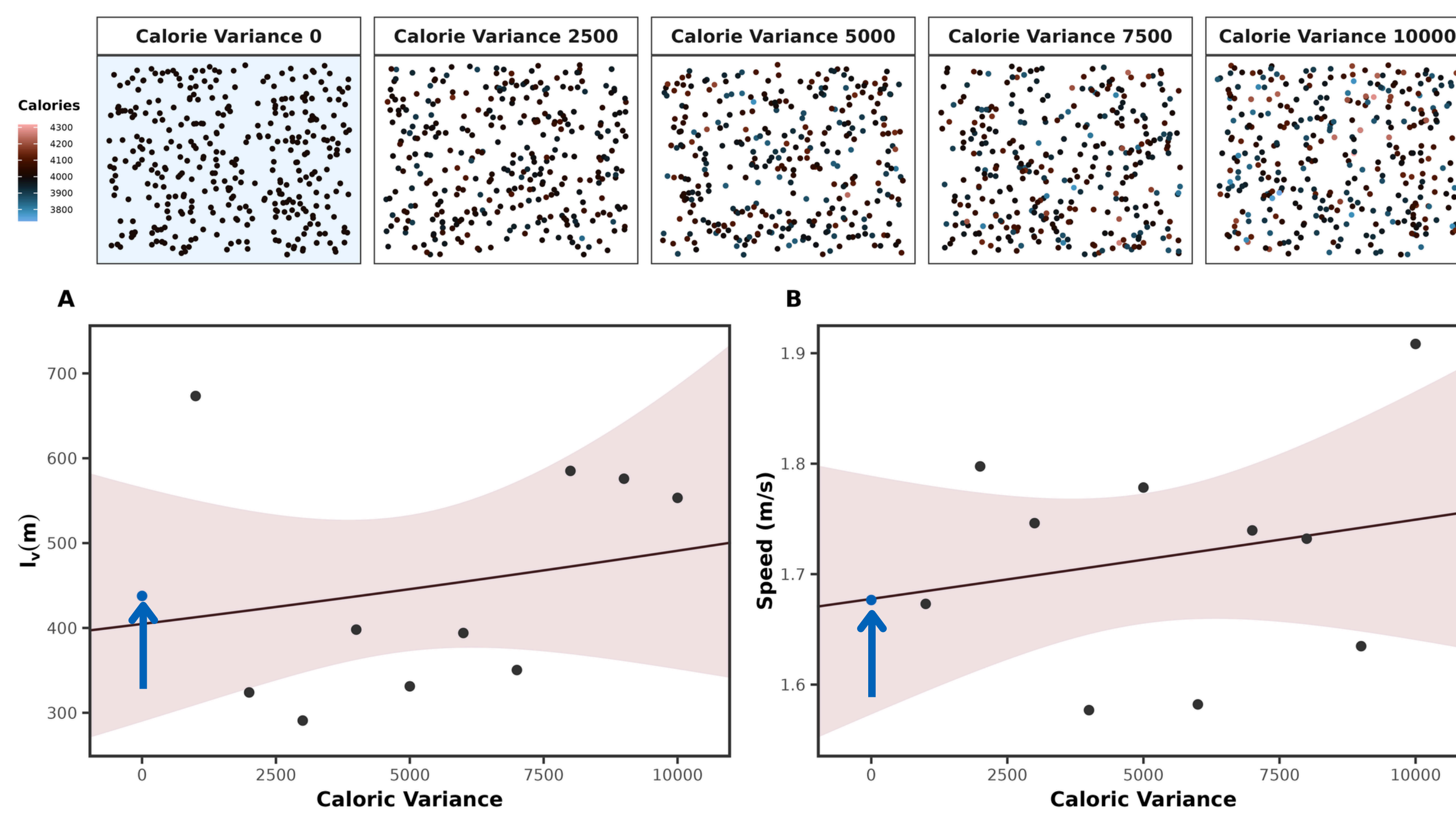


Figure 4: ESS in landscapes with increasing caloric variance. (A) Ballistic length scale (l_v) shows an insignificant increase with caloric variance (Gamma GLM, log link: $\beta = 1.9 \times 10^{-5}$, $p = 0.52$). (B) Similarly, speed increases insignificantly with caloric variance (Gamma GLM, log link: $\beta = 4.2 \times 10^{-6}$, $p = 0.47$). Blue arrow indicates parametrization that is consistent across all scenarios.